Aditya V

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Aditya V Aditya-Venkat

Education

BS **Indian Institute of Technology Kanpur**, Statistics and Data Science — 2022 – 2026
- **Awards:** Academic Excellence Award (2022 & 2023)
- **Honors Track Student**
- **Minor:** Machine Learning and Applications

CPI: 9.4/10

Grade XII **National Centre for Excellence, Bengaluru** — **Score:** 97.2% | 2022

Grade X **National Centre for Excellence, Bengaluru** — **Score:** 97.2% | 2020

Scholastic Achievements

- Received the **Academic Excellence Award, IIT Kanpur**- 2 years for excellent academic performance *2023,2024*
- Secured **All India Rank 817** in JEE Advanced & **All India Rank 721** in JEE Mains among 1 Million candidates *2022*
- Secured **All India Rank 3** in IISER Aptitude test amongst 20 thousand candidates from across the country *2022*
- Secured **All India Rank 641** in KVPY SA amongst 50 thousand candidates and became eligible for scholarship *2021*
- Secured **Karnataka State Rank** of **28** in KCET amongst 200 thousand candidates from across the state *2022*
- Secured **Karnataka State Rank** of **33** in COMEDK amongst 60 thousand candidates from across the state *2022*

Work Experience

Organizational restructuring case at global Tier 1 IT firm | Associate Consulting Intern, Bain & Co. | Mumbai (Onsite) (*May'25- Jul '25*)
- Designed and implemented **automated models** to recommend organizational restructuring actions, driving measurable margin improvements across **40+ client accounts**
- Developed and applied advanced **quantitative techniques** to estimate value creation potential ("size of prize") in restructuring exercises under limited data availability
- Optimized account-level overheads by **segmenting 40+ accounts into five clusters**, identifying cost-driving roles, and benchmarking against best-in-class cost structures

Machine Learning and Cancer Therapy Internship | TenSixty Bio | Massachusetts (remote) (*Jun '24 - Jul '24*)
- Worked in assistance to the **Stanford graduate co-founders** to develop antibody therapy drugs for **Lung Cancer**
- Developed **Data Mining**, **clustering** and **CV** methods to quantify the presence of **160,000+** antibodies
- Created **XML Parsing** scripts to mine useful information from the Uniprotk human database of **80,000+** genes
- Set up a pipeline using **NextFlow** , **Image segmentation** and **CV** to measure antibody expression in samples
- Set up a process hosted on **Amazon Web Services** to extract and analyze antibody expression across samples
- Annotated **5+** key features of **2000+** candidate genes to identify **800+** Candidate proteins for antibody therapy

Key Projects

Analysis of Spatial Transcriptomics Methodologies | Prof. Hamim Zafar | IIT Kanpur (*Sep '23 - present*)
Second-authored a paper, poster accepted to RECOMB 2025, Seoul
- Built pipelines of **10+** different Spatial Deconvolution methodologies and **10+**Spatial Domain detection methods
- Evaluating the methods on **35+** simulated datasets from **4+** different categories on over **80 Gigabytes** of data
- Designed **novel statistics** called **Bivariate Geary's C** and **simulation methods** for **imaging-based data**
- Developed essential, consistent insights of method performance using **10+** novel, standard and spatial metrics
- Insight into dataset-type specific information aiding further research by shedding light on efficient pipelines

Bayesian Bivariate Load-sharing Model | Prof. Debasis Kundu | IIT Kanpur | IIT Kanpur (*Jan '25 - present*)

- Implemented a **bivariate load-sharing model**, overcoming flaws in traditional models and testing estimation approaches on simulated datasets
- Created a novel **Bayesian extension** of the model for a variety of common priors and models for Load-Sharing models
- Implemented **MCMC sampling techniques** to estimate **credible intervals** of parameters of interest in the load-sharing models

Sampling-based Bayesian Inference Unlearning | MTH707 : Course Project | IIT Kanpur | (*Jan '25 - Apr '25*)

- Investigated **MCMC-based machine unlearning** algorithms for **Bayesian inference models** without full retraining
- Explored **KL-divergence** based inference unlearning and potential for a novel extension to **JSD-based unlearning**
- Designed methods using **low-rank Hessian approximations** and second-order information to improve unlearning accuracy by integrating **Stochastic Gradient Reimann Langevin Dynamics** with KL-divergence based unlearning
- Conducted **simulation studies** validating posterior recovery, and error bounds under data removal scenarios

Landing on the Moon | Project Mentor | BCS, IIT Kanpur (*Nov '23 - Feb '24*)

- Developed and trained a **Deep Q-Network (DQN) model** to accurately land on rough surfaces across environments
- Provided comprehensive training on Python, **Machine Learning** principles, and **Reinforcement Learning**
- Delivered in-depth lessons on the mathematical foundations of **Q-Learning** and practical guidance on training Q-learning models
- Supervised the development and benchmarking of virtual Lunar Rovers using DQNs and other solutions

Sound Refine | Project Mentor | BCS, IIT Kanpur | (*May '24 - July '24*)

- Engineered a pipeline for **noise reduction**, **transcription**, and **summarization** trained on data built via audio mixing
- Led mentees through the basics of Python, Machine Learning, Audio processing, **CNNs**, **U-Nets**, and **Transformers**
- Developed a robust audio classification model leveraging **Mel Spectrograms** and CNNs for precise **categorization**

Technical Skills

Languages: Python, C, C++, R, SQL **Other Tools:** Git, Markdown, AWS, NextFlow, RShiny, LaTeX
Libraries: Numpy, Pandas, Scikit-Learn, Matplotlib, Seaborn, HuggingFace, PyTorch, OpenCV, XML.eTree

Relevant Course Work

⋆⋆: Outstanding Performance | ⋆: Excellent Performance | † Ongoing Courses

Data Structure and Algorithms	Data Science Labs *	Ordinary Differential Equations *
Introduction to Probability Theory	Matrix Algebra and Linear Estimation	Application of Several Variable Calculus *
Theory of Statistics	Fundamentals of Computing *	Stochastic Processes *
Linear Regression and ANOVA *	Time Series Analysis *	Non-Linear Regression **
Introduction to ML **	An Introduction to Bayesian Analysis *	Markov Chain Monte Carlo *

Positions of Responsibility

Coordinator |Debating Society, IITK (*Apr'24 - Apr'25*)

- Initiated and convened the first and second editions of IITKPD with **80+** participants, managing a budget of **INR 2,50,000**
- Organised **UPDL'24** at IIT Kanpur, an national **3-tier** tournament with **3000+** participants from over **8** states
- Improved debate training quality by introducing external debate coaches and brought in funds over **INR 1,00,000**